Swarm Witness: Executable Evidence Auditing for Agent Authorization Claims

Tracks: 3 (primary — Strengthening Agent Infrastructure); 4 (application surface only)

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Abstract

When agent-safety controls produce apparently valid evidence, what additional observations, assumptions, and comparisons are required before we can justify claims about their behavior? We introduce Swarm Witness—an executable evidence-auditing method for agent authorization claims—built around source-pinned experiments, conventional reference monitors, and fail-closed findings. Classical authorization is the null hypothesis. We confirm two prior safety findings (R1 pinned SoundGate re-execution; R2 published-harness confirmation at a later stress revision), report a bounded native BMO ledger match (R3-L1) beside an incomplete decision-model comparison (R3-D unresolved), and exercise a three-way evidence classifier on synthetic and held-out artifacts (R4/R5). Findings separate conclusion from evaluation method. The method also separates comparison divergence from enforcement failure, and records evidence / decision / effect diagnoses so misbound observations and wrong decisions prevented at the effect boundary are not conflated. Primary track: strengthening agent infrastructure (Track 3); synthetic multi-principal cases illustrate a Track 4 application surface, not live swarm oversight or unbounded population safety.

1. Introduction

Research question. When agent-safety controls produce apparently valid evidence, what additional observations, assumptions, and comparisons are required before we can justify claims about their behavior?

Authorization gates, receipt graphs, and retained grants are agent-facing controls that auditors must score carefully. A mechanism can work as implemented while the evidence used to justify broader claims remains insufficient. Swarm Witness—an executable evidence-auditing method for agent authorization claims—makes that distinction testable through pinned reproductions, matched classical controls, and explicit observation boundaries. Classical authorization is the **null hypothesis**—not a new architecture we propose, and not a claim that BMO is equivalent to classical security.

Threat / failure modes. (i) An admission gate forgets an acknowledged rejection across restart when the durable log is not recovered before admit. (ii) A verifier treats possession of a copied receipt token as event attribution without exclusivity or authenticity. (iii) Retained start/grant metadata or an old receipt is treated as standing authority without consulting current grant state.

Main contributions (claimed).

1. **Executable claim qualification** that refuses to upgrade conclusions without necessary observations, and that separates *what* was established (conclusion) from *how* it was obtained (evaluation method).

2. **Pinned / harness confirmations** of two prior projects (R1/R2) as controlled exemplars—not weekend discoveries of those mechanisms.
3. **Two distinct R3 protocols: R3-D** decision-model comparison remains unresolved (missing isomorphic BMO observations); **R3-L1** reports a bounded native ledger match under fail-closed qualification. Cross-arm *divergence* refutes equality without diagnosing enforcement failure.
4. **Three-way conclusion projection (R4/R5)** with separable diagnoses (`evidence_status`, `decision_status`, `effect_status`, `failure_origin`), including misbound and held-out misleading artifacts, bounded handoff, and sequential multi-principal transfer (A/B/C)—not concurrent live swarm oversight.

Primary track: 3 (Strengthening Agent Infrastructure). Track **4** (Multi-Agent Oversight) is an application surface illustrated by synthetic multi-principal cases, not a demonstrated real-swarm result. Preexisting BMO code and Oct 25 scaffold are marked in `PREEXISTING.md`.

2. Related Work

We build on Apart Swarm Alignment themes of identity, durable commitments, and oversight under copy/rename/relay (Hammond et al., 2025; Chan et al., 2025; Greenblatt et al., 2024). Concrete targets:

- **SoundGate / "A Restart Is Not Permission"** — durable auth decisions and recovery-barrier repair (pinned package `soundgate==0.1.3`, source revision `6568403`). Original authors already reported the core repair; we re-execute under pin.
- **EvidenceGraph / Junejo "Receipt Possession Is Not Event Attribution"** — later stress harness at git rev `a5ea5c9`; paper baseline rev `efbd419` predates the stress script—we confirm the later harness, not the earlier tree.
- **BMO ordinary-retry retained grant** — preexisting durable grant / owner-witness surfaces under pinned `git_rev`; our new work is the matched ClassicalLedgerGate comparison and evidence packaging, not BMO itself.

In-repo baseline cards state inherited mechanism classes (recover-before-admit; no-implicit-exclusivity; durable grant-table checks) without equating every proxy to a decision match. Swarm Witness differs from building a new gate: it asks what labels survive matched baselines and discriminating checks.

3. Methods

Claim cards (`claims/*.yaml`) fix `property`, `scope`, `quantifier`, `success_criterion`, and `assumptions`. **Pins** freeze revisions. **Findings** carry a conclusion label and an `evaluation_method`. `scripts/check_evidence.py` fails closed if the index disagrees with findings; it does not re-execute harnesses.

R1. Pinned SoundGate re-execution at `6568403` (`evaluation_method=pinned_reexecution`).

R2. Confirmation of the authors' later-published stress harness at `a5ea5c9` (`published_harness_confirmation`)—not execution of paper rev `efbd419`.

R3-D. Decision-model comparison; unmatched BMO observations → `unresolved` (comparison not established).

R3-L1. Native ledger match at pin `4b717d7`. Disagreement → `comparison_divergent` (equality refuted ≠ `failed_enforcement`).

R4. Three-way projection plus diagnoses; misbound → `failure_origin=evidence`; authentic broken gate → `decision_policy`; held-out fixture under

`experiments/evidence_sufficiency/held_out/`.

R5. Sequential A/B/C model; broken allow → `effect_status=prevented`, `failure_origin=decision_policy`.

4. Results

Claim	Conclusion	Evaluation method
R1 SoundGate	`reproduced`	`pinned_reexecution`
R2 EvidenceGraph	`reproduced`	`published_harness_confirmation`
R3-D decision-model	`unresolved`	`matched_decision_model`
R3-L1 ledger	`classical_baseline_matches`	`native_matched_baseline`
R4 evidence sufficiency	`reproduced`	`synthetic_protocol_conformance` (+ held-out)
R5 multi-principal	`reproduced`	`synthetic_protocol_conformance`

Metrics cite `evidence/*/summary.json`.

4.1 R1 — SoundGate recovery barrier (`reproduced`)

Fresh local rebuild+sweep; `all_checks_pass: true`; `raw_sha256=51b4acd6…`. Histories per policy: **153**.

Policy	cases	prohibited_effect	refused_before_ack	supported	fresh_work
original	153	**143**	4	6	149
strict	153	0	**147**	6	6
barrier	153	**0**	0	**153**	153

Barrier restores acknowledge-and-restrict across restart under the published histories; original loses restrictions (143 prohibited effects). This establishes reproducibility within tested conditions, not a new recovery principle.

4.2 R2 — Receipt ≠ attribution (`reproduced` = published-harness confirmation)

Later stress harness at `a5ea5c9` (not paper rev `efbd419`); `condition_by_seed_checks` **36**, passed **36**; seeds 0,1,2; `all_contract_checks_pass: true`; `source_tree_sha256=2962568f…`.

Metric (per seed)	seed 0	seed 1	seed 2
old_rule wrong attributions on relay	1	1	1
safer supported on relay	0	0	0
safer correct on relay	0	0	0
clean withheld without exclusivity	12	12	12
shared_tokens correct	0	0	0
false_exclusivity confident errors	1	1	1

4.3 R3-D — Decision-model comparison (`unresolved`)

0/4 isomorphic BMO observations; Finding `r3-f-comparison` **`unresolved`** — comparison **not established**, not an observed negative equality result.

Case	ClassicalGate	BMO decision	BMO status
retained_start_claim_relaunch	deny	—	missing
rejected_execute	deny	—	missing
active_grant_execute	allow	—	missing
stale_grant_version	deny	—	missing

4.4 R3-L1 — Ledger claim (`classical_baseline_matches`)

Across T01–T06, ClassicalLedgerGate and native BMO L1 at pin `4b717d7` matched dispositions and projections with both arms satisfying declared properties. Equality failure would be `comparison_divergent`, not automatic `failed_enforcement`. Selected run: `evidence/r3/ledger_claim_v1/2026-11-07-ledger-v1-repair2/`.

4.5 R4 — Three-way projection (`reproduced` = synthetic + held-out)

Cases under `r4-evidence-sufficiency-v1` exercise all three classes, bounded handoff, misbound controls (`failure_origin=evidence`), authentic broken gates (`failure_origin=decision_policy`), and held-out `held_out_misleading_grant_observation`. Finding `r4-f-evidence-sufficiency` **`reproduced`**; `all_checks_pass: true`.

4.6 R5 — Multi-principal transfer (`reproduced` = synthetic conformance)

Ten sequential A/B/C cases: missing observation (2); wrong-decision allows with `effect_status=prevented` (2); supported contention, assignee hops, and ledger-bound receipt deny (6). Finding `r5-f-swarm-audit` **`reproduced`**; `model: sequential_shared_grant_multi_principal`.

5. Discussion

Synthesis. An executable audit can preserve the difference between an observed security property, an incomplete comparison, comparison divergence, and a conclusion invalidated by unreliable evidence.

Implications. Diagnostic labels must not introduce the unsupported inferences the method was built to catch: equality ≠ enforcement; misbinding ≠ gate policy; wrong decision with prevented effect ≠ committed illicit effect.

Limitations (summary). R3-D incomplete; R3-L1 projection-scoped; R4/R5 mostly synthetic conformance (one held-out fixture); no live concurrent agents. Full Limitations appendix follows.

Future work. Isomorphic R3-D probes; more externally supplied artifacts; genuine contested concurrent handoffs.

Appendix A. Limitations and Dual-Use / Ethical Considerations

What each result does and does not establish

R1 (`reproduced`). Establishes that, on the pinned SoundGate recovery-barrier harness (source revision `6568403`), the barrier policy yields `prohibited_effects = 0` across 153 histories while the original policy yields 143 prohibited effects, with 147 strict refusal-startup refusals under the strict policy. Does **not** establish that every agent admission gate is restart-safe, that SoundGate is free of other defect classes, or that recover-before-admit is a new multi-agent invention (that classical reduction remains an authored baseline question under test).

Committed `evidence/r1/raw.json.gz` supports local inspection of the sweep artifact (`gzip -dc evidence/r1/raw.json.gz > evidence/r1/raw.json`); recorded `raw_sha256` is an integrity check and **does not replace** inspecting `raw.json` / analysis case records.

R2 (`reproduced`). Establishes that the stress pack executed at harness rev `a5ea5c9` passes 36/36 condition-by-seed checks: old-rule relay wrong attributions are 1 per seed; safer relay supported attributions are 0; clean runs withhold all 12 attributions without exclusivity declarations. Paper baseline `efbd419` is the pre-extension tree (no `incident_stress.py`); its baseline *behavior* is represented by replaying the authors' later-published stress pack at `a5ea5c9`, not by running a missing harness at `efbd419`. Does **not** establish that all receipt or ledger systems are safe under relay/copy, or that exclusivity declarations themselves cannot be gamed outside the modeled threat surface.

R3 decision-model (`unresolved`). Establishes that the matched ClassicalGate decision-model comparison against BMO is **incomplete**: 0/4 required cases have isomorphic BMO observations, so the comparison Finding is `unresolved` (comparison not performed). ClassicalGate is a **decision-model** null (`state`, `request`, `grant_version`)—not a restart-durability claim. Selected preferred Go tests on the pinned BMO revision are package metadata only and do not fill decisions or substitute for a matched comparative label. Does **not** establish ClassicalGate ↔ BMO API equivalence, assumption-dependent affinity, or a methodological reduction of BMO's durable-grant story to classical controls.

R3 ledger-claim-v1 (`classical_baseline_matches`, bounded). Establishes equal dispositions and property-relevant projections (grant_spent, counts, call_id sets) across T01–T06 for ClassicalLedgerGate vs native `ClaimOrdinaryRetainedGrantFirstEditTx` at pin `4b717d7` under fail-closed native qualification. Does **not** claim synthetic classical grant/epoch aliases equal native durable hashes, full event-trace observational equivalence, live execution safety, process-restart correctness, concurrency behavior, or general BMO equivalence. Legacy decision-model R3 remains separately unresolved.

R4 three-way projection (`reproduced` = synthetic conformance + held-out). Establishes classification into missing_evidence / failed_enforcement / supported with separable diagnoses, including misbound evidence (`failure_origin=evidence`) and one held-out misleading artifact. Does **not** prove the taxonomy is necessary or sufficient versus other assurance models. `reproduced` here is protocol conformance, not external replication.

R5 multi-principal transfer (`reproduced` = synthetic conformance). Establishes taxonomy transfer on a sequential three-principal model; broken allows use `effect_status=prevented` and `failure_origin=decision_policy`. Does **not** establish concurrent worker execution or live swarm oversight.

Finite experiments and checker scope

Pinned histories, seeds `{0,1,2}`, and preferred Go tests are finite. Passing them does not imply equivalence across versions, storage backends, concurrent writers outside the model, or production deployments. Labels apply to the stated scope on the claim cards only. `scripts/check_evidence.py` binds reported metrics to linked Run observations when keys overlap; it does not re-execute harnesses or prove metric completeness beyond that binding.

Dual-use / adversarial use of the audit method

An executable claim-audit method can help **defenders** find weak gates and attribution gaps—and can also help **adversaries** locate the same gaps (restart races, possession-as-attribution, retained-metadata relaunch). We publish **tests, pins, matched baselines, and labels**, not

exploit kits, weaponized harnesses, or production credentials. Reproduction artifacts are scoped to already-public or author-local preexisting trees (`PREEXISTING.md`).

Preexisting code vs weekend analysis

Oct 25 scaffold paths and the external BMO tree at pinned `bmo_root` / `git_rev` predate this sprint and were not modified for R3 beyond read-only `go test`. Weekend work is the Swarm Witness schema, pins, claim cards, baselines, experiment wrappers, evidence records, index checker, and this report. Judges should score the audit contribution, not treat preexisting BMO surfaces as sprint inventions.